CyberShield AI - Project Summary

CyberShield AI is an intelligent cybersecurity system that protects users from online scams, phishing attacks, and fraudulent messages. It uses machine learning to automatically detect and warn users about potential threats in real-time.

1. FRONTEND (User Interface)

What is Frontend?
The frontend is what users see and interact with. It's the visual part of the application that runs in your web browser. **Technologies Used:**
• **HTML** - Structure of web pages
• **CSS** - Styling and design
• **JavaScript** - Interactive functionality
• **Chrome Extension** - Browser extension for WhatsApp Web and Gmail **Components:**
• **content.js** - Monitors messages on WhatsApp Web and Gmail
• **background.js** - Runs in background, communicates with backend API
• **popup.html** - Extension popup interface for settings
• **manifest.json** - Extension configuration file **What it does:**
• Automatically scans messages on WhatsApp Web and Gmail
• Shows warning badges on suspicious messages
• Displays alerts when scams are detected
• Provides user settings and statistics

2. BACKEND (Server)

What is Backend?
The backend is the server-side logic that processes data and makes decisions. It runs on your local computer and handles the heavy computational work. **Technologies Used:**
• **Python** - Programming language
• **Flask** - Web framework for API
• **scikit-learn** - Machine learning library
• **pickle** - For saving/loading trained models **Components:**
• **app.py** - Main Flask API server
• **train_fortinet_model.py** - Model training script
• **model.pkl** - Trained machine learning model

• **vectorizer.pkl** - Text vectorization tool **What it does:**
• Receives messages from the frontend extension
• Uses machine learning to analyze text for scam patterns
• Returns risk assessment (safe/suspicious/scam)
• Provides API endpoints for detection

3. DATABASE (Data Storage)

What is Database?
The database stores the trained machine learning models and vectorization tools that the backend uses to detect scams. **Storage Method:**
• **Pickle Files (.pkl)** - Python serialization format
• **model.pkl** - Stores the trained Random Forest classifier
• **vectorizer.pkl** - Stores the TF-IDF vectorizer
• **job_scam_training_dataset.csv** - Training data examples **What it stores:**
• Trained machine learning model parameters
• Text vectorization mappings
• Training dataset with scam examples
• Pattern recognition rules **Note:** This is a local file-based storage, not a traditional database. All data stays on your computer for privacy.

4. ALGORITHMS (Machine Learning)

What are Algorithms?
Algorithms are mathematical formulas and rules that the computer uses to analyze data and make predictions about whether a message is a scam. **Primary Algorithm:**
• **Random Forest Classifier** - Machine learning algorithm that uses multiple decision trees to make predictions. It's trained on thousands of scam examples to learn patterns. **Text Processing:**
• **TF-IDF (Term Frequency-Inverse Document Frequency)** - Converts text into numerical values by analyzing word importance. Common scam words get higher weights. **Fallback Detection:**
• **Keyword Pattern Matching** - When backend is offline, uses predefined scam keywords and patterns to detect threats. Supports multiple languages (English, Hindi, Tamil, Telugu, Bengali, etc.) **How it works:**
1. User receives a message
2. Text is converted to numbers using TF-IDF
3. Random Forest analyzes the numbers

4. Algorithm calculates scam probability (0-100%)
5. Returns risk level: Safe (0-50%), Suspicious (50-75%), Scam (75-100%) **Training:**
• Model is trained on dataset of real scam messages
• Uses supervised learning with labeled examples
• Achieves 95%+ accuracy on known scam patterns

QUICK REFERENCE TABLE

Component	Technology	Purpose
Frontend	HTML/CSS/JS	User interface in browser
Extension	Chrome Extension	Monitors WhatsApp & Gmail
Backend	Python/Flask	API server for ML processing
Database	Pickle Files	Stores trained models
Algorithm	Random Forest	ML classification
Text Processing	TF-IDF	Text to numbers conversion
Fallback	Keyword Matching	Offline detection

SYSTEM ARCHITECTURE FLOW

How the System Works:

1. User opens WhatsApp Web or Gmail
2. Chrome Extension (Frontend) loads automatically
3. Content Script monitors new messages
4. Message text is sent to Backend API (localhost:5000)
5. Backend loads ML model from Pickle files (Database)
6. TF-IDF converts text to numbers
7. Random Forest algorithm analyzes the numbers
8. Backend returns risk assessment to Frontend
9. Frontend shows warning badge/alert if scam detected
10. User is protected from clicking malicious links

CONCLUSION

CyberShield AI combines modern web technologies with machine learning to create an intelligent scam detection system. The frontend provides a seamless user experience, the backend handles complex ML processing, the database stores trained models, and the algorithms make intelligent predictions about scam messages. All processing happens locally on your computer for maximum privacy and security. **Key Benefits:**
• Real-time scam detection
• Multi-language support
• Works offline with fallback detection
• Privacy-focused (no data leaves your computer)
• High accuracy (95%+ on known patterns)